ALGORITHMS ANALYSIS AND DESIGN

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Computer Applications

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ALGORITHMS ANALYSIS AND DESIGN

Introduction, Analysis Framework and Sorting Techniques

Sorting Algorithms – Divide and Conquer

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ALGORITHMS ANALYSIS AND DESIGN
Quick sort

- It was invented by a British scientist C.A.R Hoare.

- Quick sort divides the elements according to their values.

Figure 2: https://en.wikipedia.org/wiki/Tony_Hoare

ALGORITHMS ANALYSIS AND DESIGN
Quick Sort

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- Quick Sort uses the Divide and Conquer paradigm to sort a given list of numbers.
- The execution of Quick Sort typically follows three key steps
  - Divide
  - Conquer
  - Combine

ALGORITHMS ANALYSIS AND DESIGN
Quick Sort - Divide

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- The algorithm starts with the division process.
  - It involves the selection of a pivot element from the array.
- The pivot element is used to divide the array into two segments.
- The aim of this division is to ensure that the array is rearranged in such a way that all elements that are less than the pivot element are placed before it and all elements that are greater than the pivot element are placed after it.

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Quick Sort - Conquer

- Once the partitioning is finished, the pivot element will be in its correct position in the array.
- Once the array is partitioned, the algorithm enters the conquer phase.
- This phase includes the two partitioned arrays created from the elements that lie before and after the pivot and the recursive application of the same Quick Sort algorithm.

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Quick Sort - Conquer

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- For each of these sub arrays, a new pivot is chosen and the sub array is partitioned around the pivot.
- The recursive application of the algorithm continues until the base case is reached.
  - This is when the sub array has only one element.

ALGORITHMS ANALYSIS AND DESIGN
Quick Sort - Combine

- Quick Sort does not require a combine step.
- It is an in-place sorting algorithm.
- It does not require the creation of a new array to hold the sorted elements but rather holds the elements in the original array itself.

ALGORITHMS ANALYSIS AND DESIGN
Quick Sort - Combine

- This means that once the conquer step has sorted the sub-arrays, the original array is already sorted.
- This means that there is no need to combine the sorted sub-arrays.

ALGORITHMS ANALYSIS AND DESIGN

Quick Sort - Methodology

- The core operation of Quick Sort lies in the selection of a pivot element from the array.
- The remainder of the array is then divided into two distinct sections based on this pivot, those elements which are less than the pivot and those which are greater.
- The selection of the pivot element is not a trivial task as it can greatly impact the efficiency of the sorting process.

ALGORITHMS ANALYSIS AND DESIGN

Quick Sort - Methodology

- An ideal pivot would divide the array into two nearly equal halves, leading to an optimal Quick Sort operation.
- Once the array has been partitioned based on the pivot, the pivot element will find its accurate place in the final sorted array.
- It separates all elements smaller than it on its left side and all elements larger on its right side.

ALGORITHMS ANALYSIS AND DESIGN
Quick Sort - Methodology

- After partitioning around the pivot, the algorithm is reapplied to each of the two partitions, elements less than the pivot and elements greater than the pivot.

- This process repeats until the sub array has one element.

$$\underbrace{A[0]\ldots A[s-1]}_{all\ are \leq A[s]}\ A[s]\ \underbrace{A[s+1]\ldots A[n-1]}_{all\ are > A[s]}$$

ALGORITHMS ANALYSIS AND DESIGN

Quick Sort - Steps

- Step 1
  - Select a pivot. This could be the first, middle, or last element of the array or a random element.
- Step 2
  - Partition the array around the pivot.
  - Move elements smaller than the pivot before it and elements larger than it after.

Quick Sort - Steps

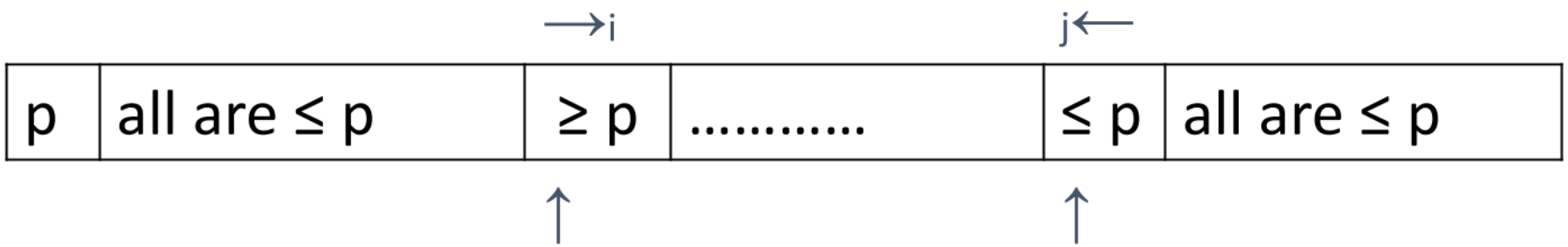
- Step 2 (Left to right scan)
  - If A[i] ≤ pivot, increment i by 1.
    - It starts with the second element.
    - It skips all elements smaller than the pivot and stops when a bigger or equal element is found.

Quick Sort - Steps

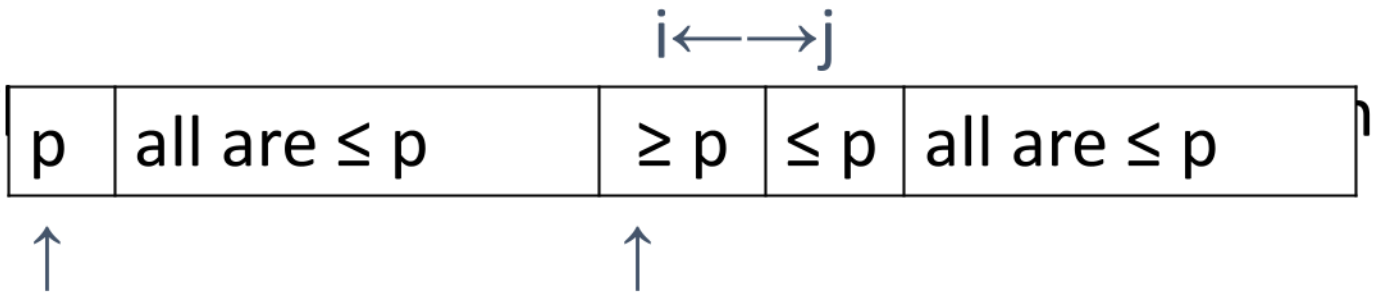
- Step 2 (Right to left scan)
  - If A[j] > pivot, decrement j by 1.
    - It starts with the last element of the sub array.
    - It skips all elements bigger than the pivot and stops when a smaller or equal element is found.

ALGORITHMS ANALYSIS AND DESIGN
Quick Sort - Steps

- Step 2 (Right to left scan)
  - If i < j
    - i and j have not crossed.
    - Exchange A[i] and A[j] and continue.



ALGORITHMS ANALYSIS AND DESIGN
Quick Sort - Steps

- Step 2 (Right to left scan)
  - If i > j
    - i and j have crossed over.
    - T 

nging the pivot with $A_{[j]}$.

Quick Sort - Steps

- Step 3
  - Repeat these steps for the sub arrays on both sides of the pivot.
  - This is a recursive operation.

ALGORITHMS ANALYSIS AND DESIGN
Algorithm – Partition(A[l..r])

- // Partitions a sub array by using its first element as a pivot.
- // Input : A Sub array A[l….r] of A[0….n-1], defined by its left and right //indices l and r (l<r).
- // Output : A partition of A[l…r], with the split position returned as this //function’s value.

ALGORITHMS ANALYSIS AND DESIGN

Algorithm – Partition(A[l..r])

p← A[l]

i ←l+1; j ← r

while (true)

repeat i ← i+1 until A[i]< p

repeat j ← j-1 until A[j] ≥ p

swap (A[i], A[j])

until i ≥ j

swap (A[i],A[j]) //undo last swap when i ≥ j

swap(A[l], A[j])

return j

ALGORITHMS ANALYSIS AND DESIGN

Algorithm – Quicksort(A[l..r])

- // Sorts a subarray by quicksort
- // Input: Subarray of array A[0..n − 1], defined by its left and right indices l

  //and r
- // Output: Subarray A[l..r] sorted in nondecreasing order

ALGORITHMS ANALYSIS AND DESIGN
Algorithm – Quicksort(A[l..r])

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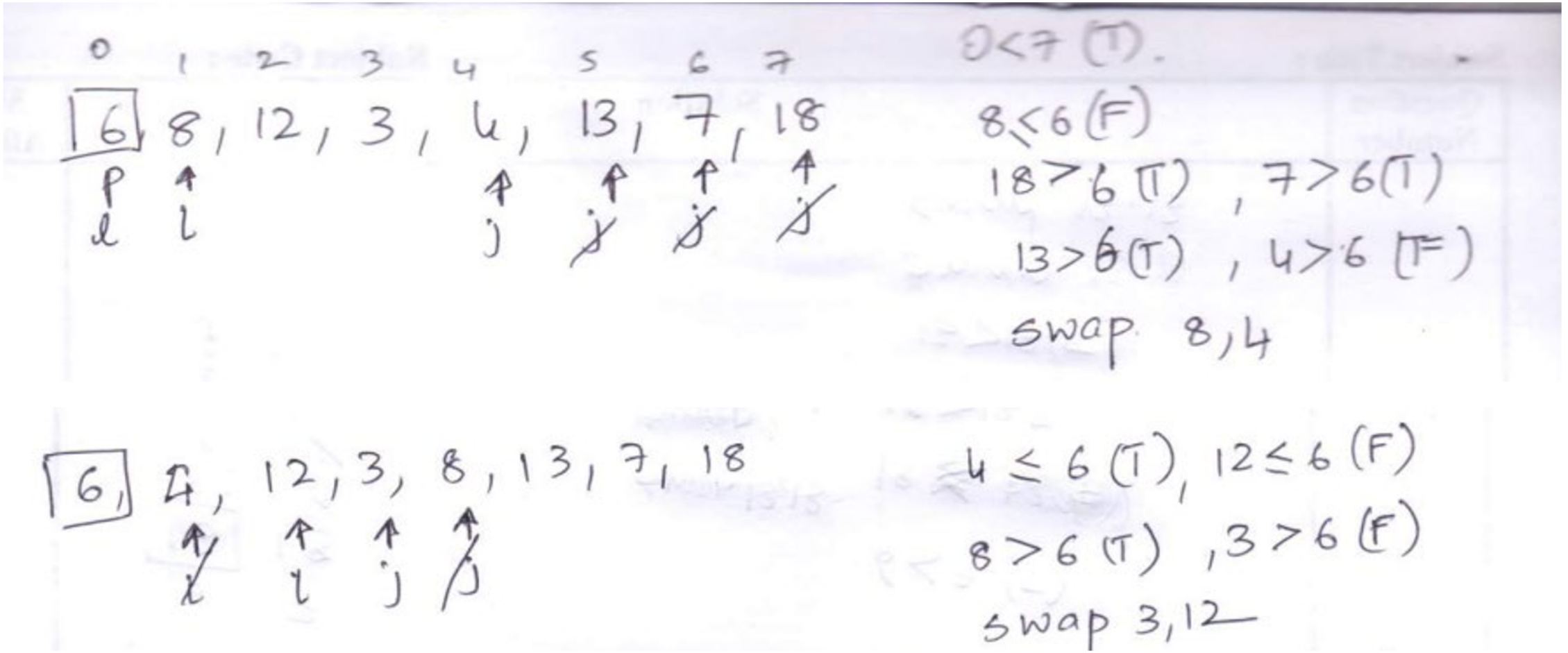
- if l < r

  s ← Partition(A[l..r]) // s is a split position

  Quicksort(A[l..r − 1])

  Quicksort(A[s + 1..r])

Algorithm – Partition(A[l..r]) - Tracing

- 

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Algorithm – Partition(A[l..r]) - Tracing

- [6], 4, 3, 12, 8, 13, 7, 18.

  i i
  j i

  $3 \leq 6$ (T), $12 \leq 6$ (F)

  $12 > 6$ (T), $3 > 6$ (F)

  swap 3, 12

  [6], 4, 12, 3, 8, 13, 7, 18.

  j i

  swap 12, 3.

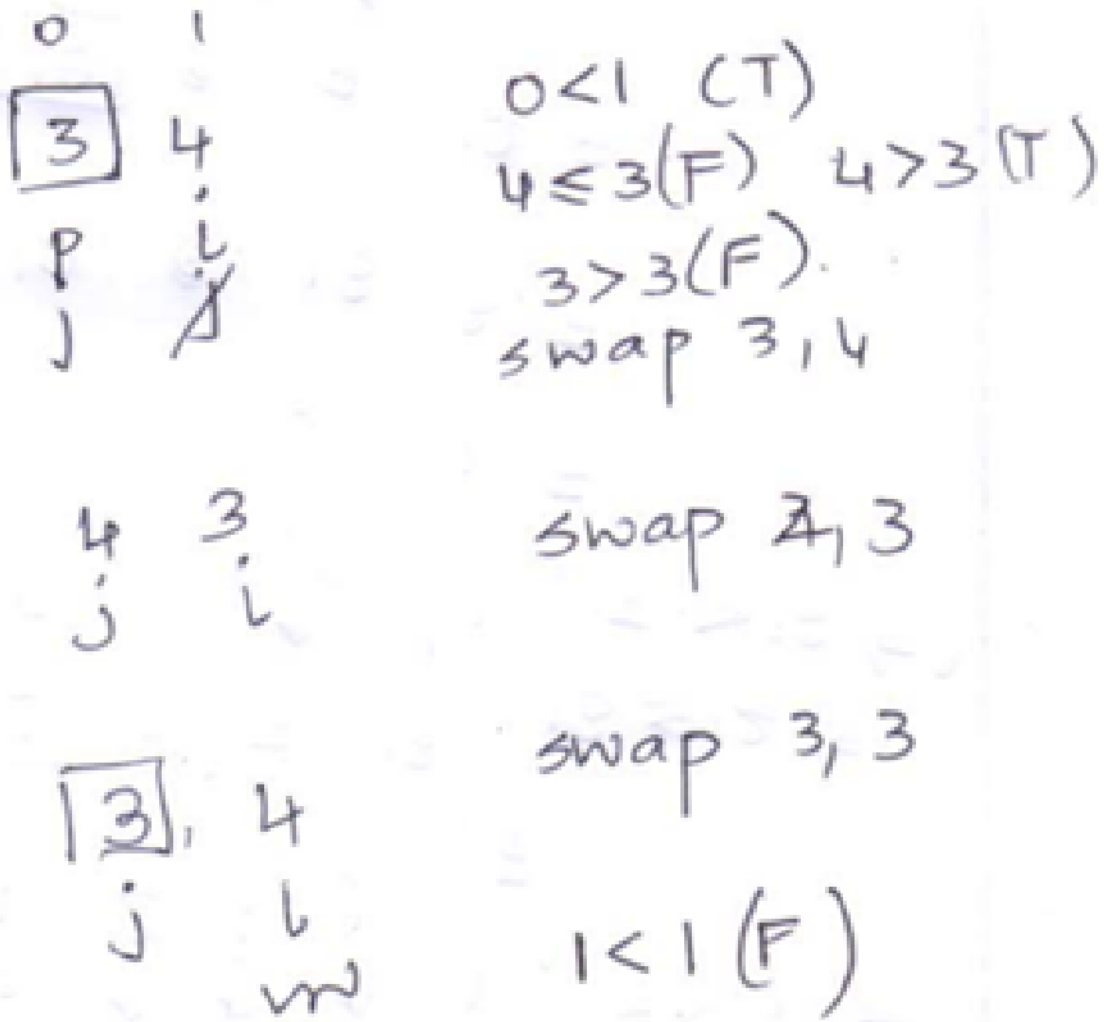
  [6], 4, 3, 12, 8, 13, 7, 18

  j i

  swap 6 and 3.

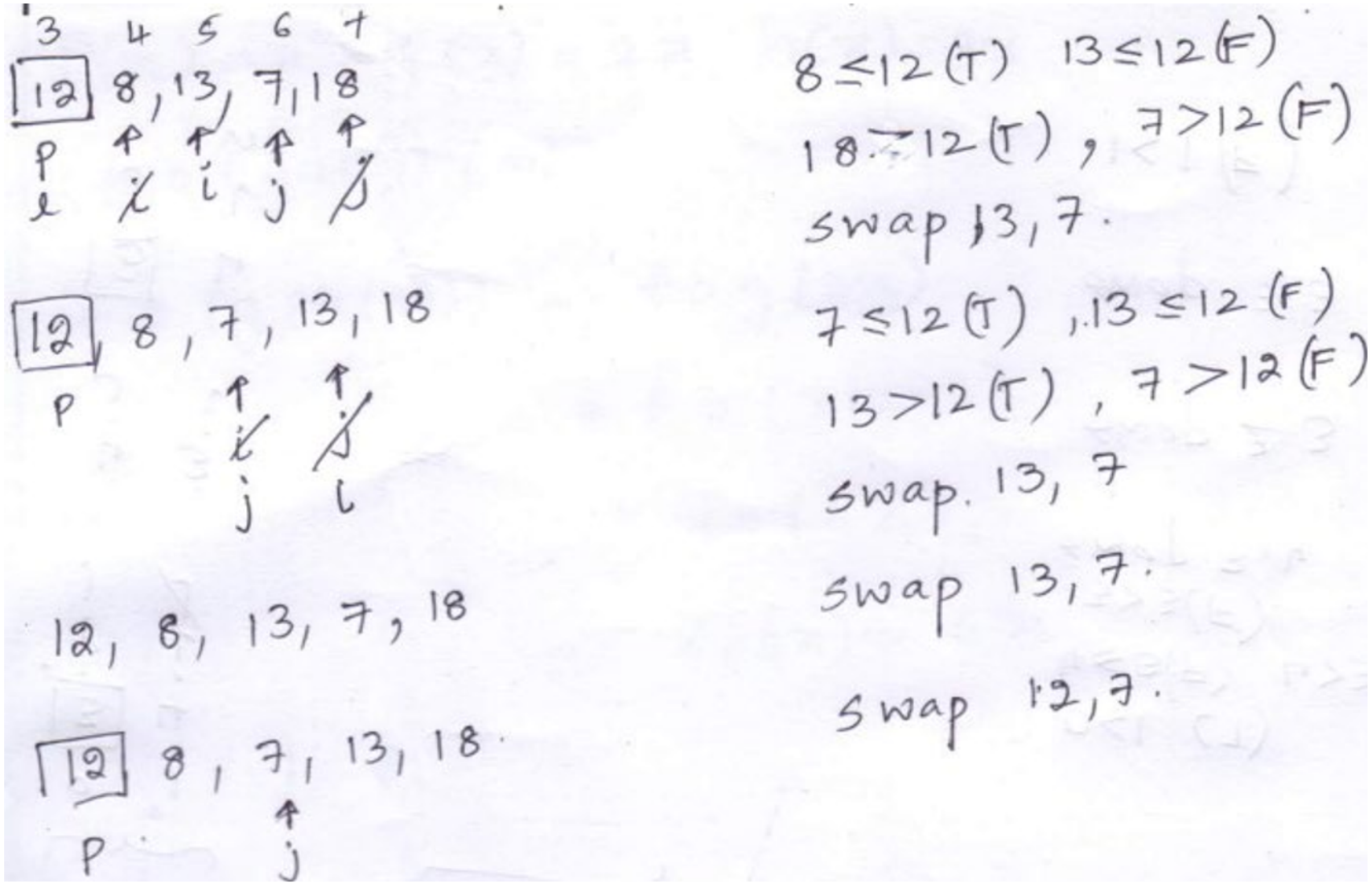
  3, 4, [6], 12, 8, 13, 7, 18

Algorithm – Partition(A[l..r]) - Tracing

- 

ALGORITHMS ANALYSIS AND DESIGN

Algorithm – Partition(A[l..r]) - Tracing

- 

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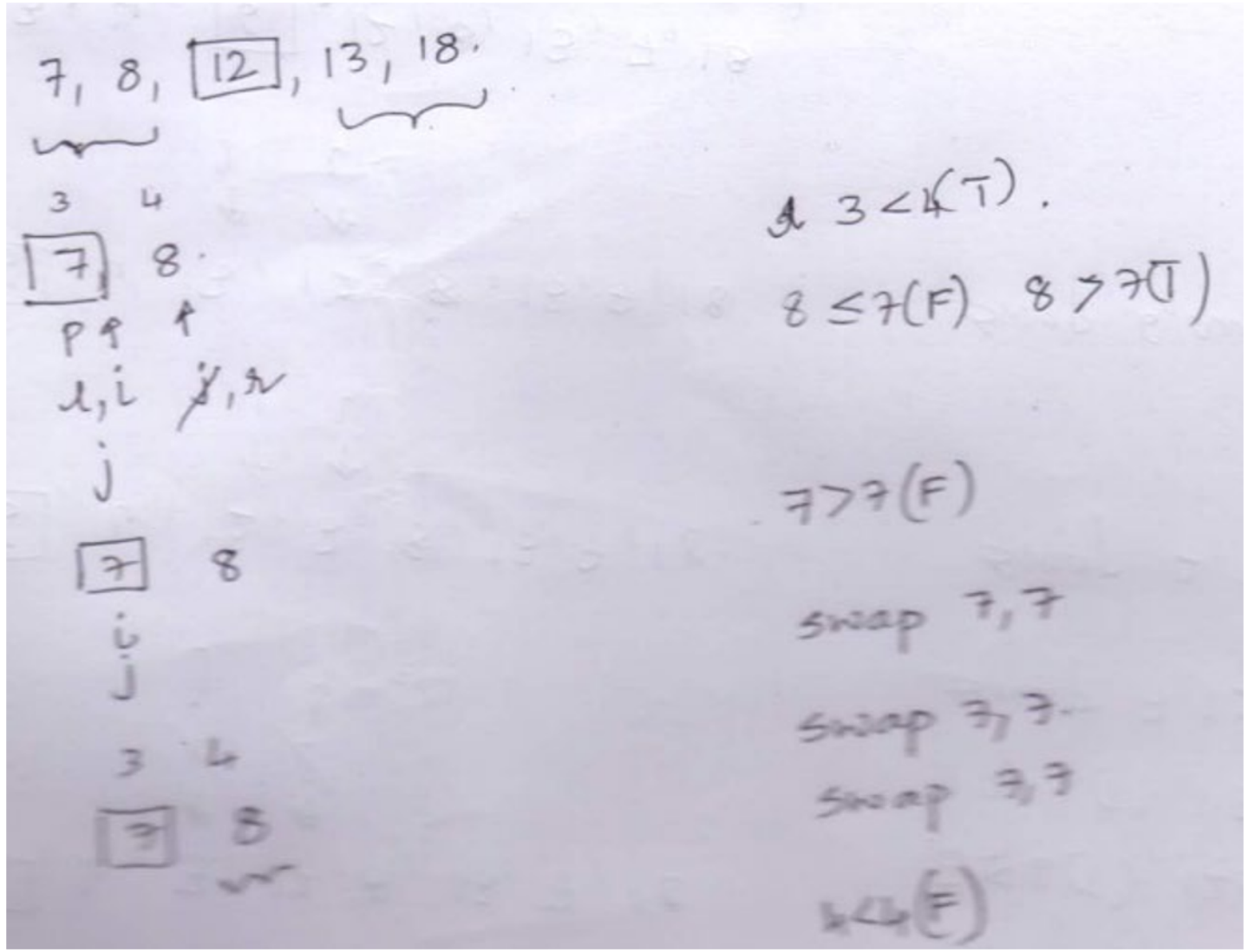
Algorithm – Partition(A[l..r]) - Tracing

- 

ALGORITHMS ANALYSIS AND DESIGN

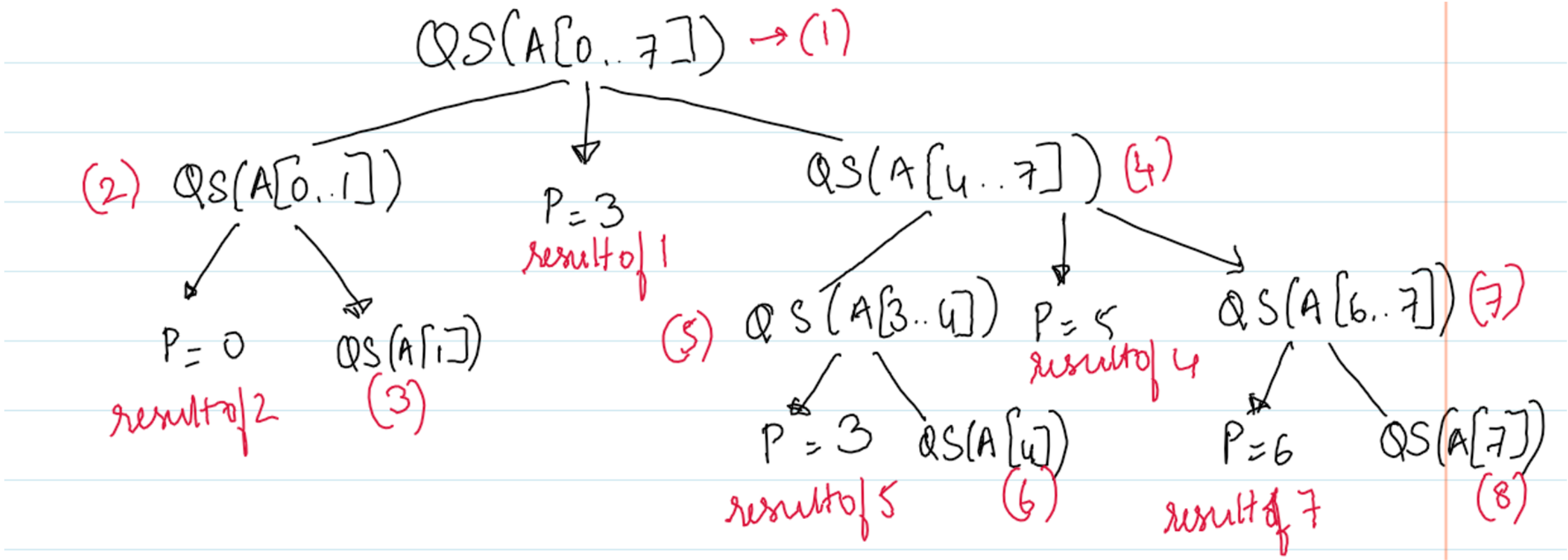
Algorithm – Partition(A[l..r]) - Tracing

- 6 7 — 6<7 (T)
  13, 18 — 18≤13(F) 18>13(T) 13>13(F)
  p, l r — Swap 13,18
  j i, ~~j~~
  18 13 — swap 13,18
  j i
  13 18 — Swap 13,13
  j i
  6 7
  13 18 — 7<7 (F)

  3, 4, 6, 7, 8, 12, 13, 18

ALGORITHMS ANALYSIS AND DESIGN

Recursive Tree



ALGORITHMS ANALYSIS AND DESIGN
Merge Sort

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- It was created by John von Neumann in 1945.

Fig 3: https://en.wikipedia.org/wiki/John_von_Neumann

ALGORITHMS ANALYSIS AND DESIGN

Merge Sort

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- It was created by John von Neumann in 1945.
- It sorts a given array A[0..n-1] by dividing it into two halves A[0..(n/2)-1] and A[n/2..n-1].
- Sort each of them recursively
- Merge the two smaller sorted arrays into a single sorted one.

ALGORITHMS ANALYSIS AND DESIGN
Algorithm – Mergesort(A[0..n-1])

- //Sorts array A[0..n-1] by recursive merge sort
- //Input: An array A[0..n-1] of orderable elements
- //Output: Array A[0..n-1] sorted in non-decreasing order

ALGORITHMS ANALYSIS AND DESIGN

Algorithm – Mergesort(A[0..n-1])

- if n > 1
- copy A[0..(n/2)-1] to B[0..(n/2)-1]
- copy A[(n/2)..n-1] to C[0..(n/2)-1]
- Mergesort(B[0..(n/2)-1])
- Mergesort(C[0..(n/2)-1])
- Merge (B,C,A)

ALGORITHMS ANALYSIS AND DESIGN
Algorithm – Mergesort(A[0..n-1]) - Methodology

- This can be done as follows
  - Two pointers (array indices) are initialized to point to the first elements of the arrays being merged.
  - Then the value of the index compared and smaller of them is added to new array being constructed.

ALGORITHMS ANALYSIS AND DESIGN
Algorithm – Mergesort(A[0..n-1]) - Methodology

- Now the index of the smaller element array is incremented to point to its immediate successor in the array it was copied from.
- This operation is continued until one of the two given arrays is exhausted.
- The elements of the remaining array will be copied to the end of the new array.

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Algorithm – Merge(B[0..P-1], C[0..Q-1],A[0..P+Q-1])

- // Merges two sorted arrays into one sorted array
- //Input: Arrays B[0..P-1] and C[0..Q-1], both sorted
- //Output: Sorted array A[0..P+Q-1] of the elements B and C

ALGORITHMS ANALYSIS AND DESIGN

Algorithm – Merge(B[0..P-1], C[0..Q-1],A[0..P+Q-1])

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```
i←0, j ←0, k ← 0

while i<p and j < q

    if B[i] ≤ C[j]

        A[k] ← B[i], i ← i+1

    else

        A[k] ← C[j], j ← j + 1

    k ← k + 1
```

ALGORITHMS ANALYSIS AND DESIGN

Algorithm – Merge(B[0..P-1], C[0..Q-1],A[0..P+Q-1])

```
if i = p
    copy C[j..q-1] to A[k..p+q-1]
else
    copy B[i..p-1] to A[k..p+q-1]
```

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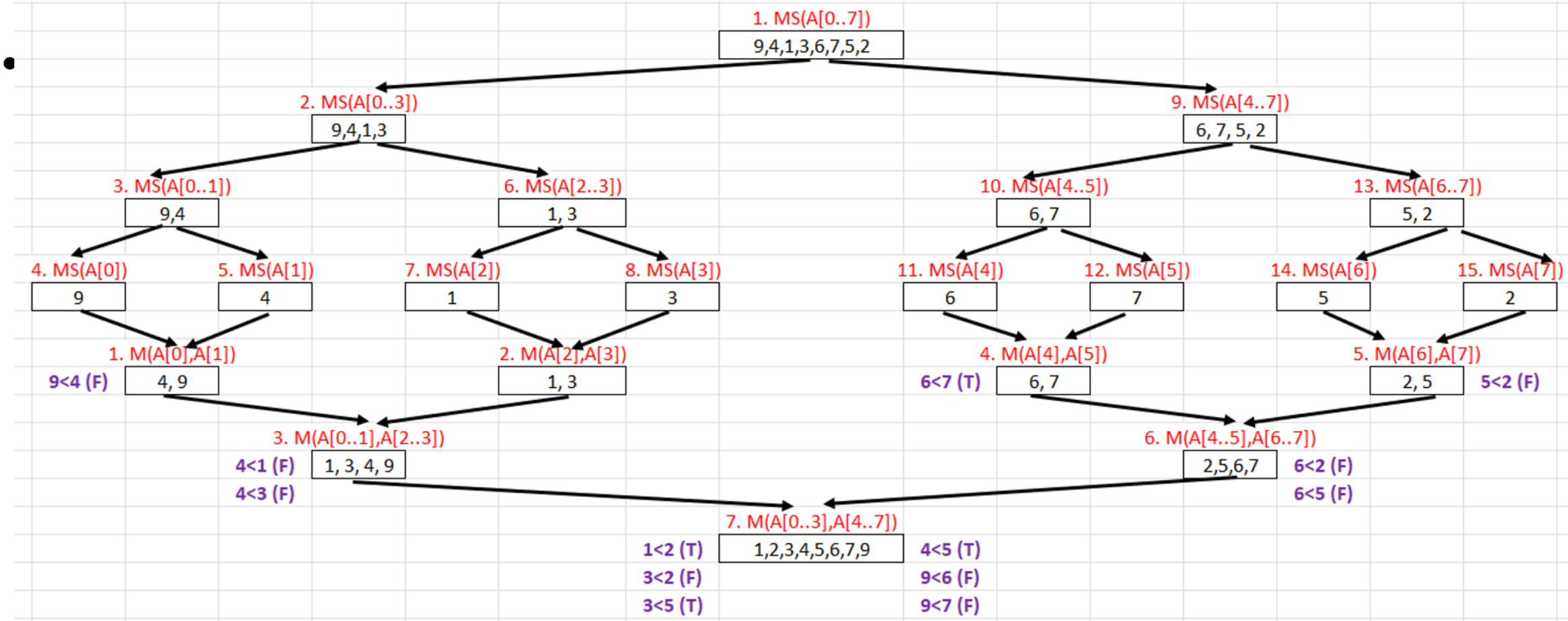
Algorithm – Merge Sort - Tracing



ALGORITHMS ANALYSIS AND DESIGN

Recap

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- Quick Sort
- Merge Sort
  - Algorithm
  - Tracing

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THANK YOU

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